

BUILD GUIDE

Bittle X Pi Add-on

A Raspberry Pi Zero 2 W with a battery, a speaker and a microphone, in one rounded box that screws to a new back cover of the Petoi Bittle X.

Add-on v0.26
2026-11-01

4 + 2 screws
M3 inserts (FDM) or M2.5 nuts (resin)

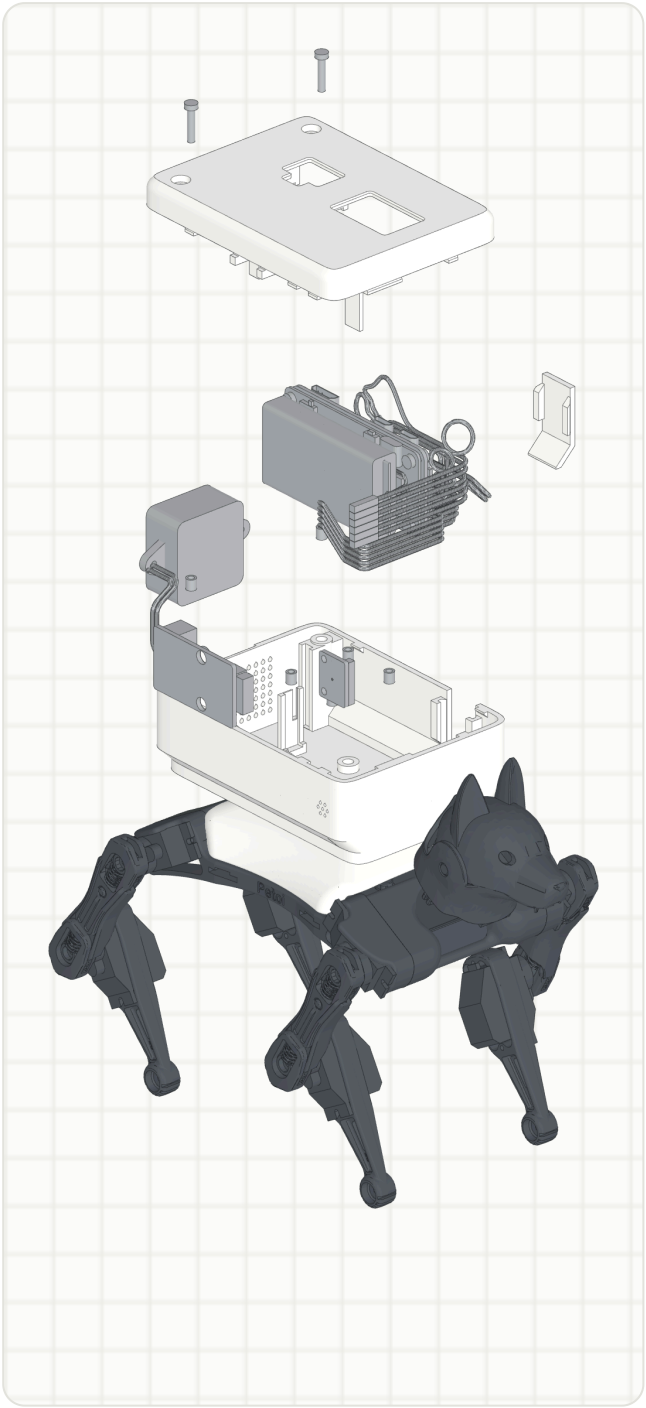
114 g
with the electronics

No supports
for the add-on parts (FDM)

OVERVIEW

What you build

The add-on is a body that screws to a cover for screws, and a slide-on top panel with two screws. The cover and the body clip onto the robot as one unit, in place of the stock cover. Every part goes into the body from the top, in the order of this guide. You need a screwdriver, a soldering iron for the splices (and the inserts of an FDM print), and a dab of glue for the amp.



Directions in this guide

Front is the head end of the robot. Left and right are the robot's left and right. The Pi plugs and the BiBoard header are in the **left well**. The amp and the mic are on the **right wall**. The speaker is at the rear.

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The 3D viewer shows every step with animation:
https://beetlebugorg.github.io/bittlex/viewer/

1

Parts to print

Print the parts in PLA, in the orientation of the files in stl/print/. Each file is for 0.20 mm layers. For 0.10 mm layers, print the *_fine file of the same name.

PART	ON THE BED	MASS
Body stl/print/pod_body_screw.3mf	floor on the bed	40 g
Top panel (second color) stl/print/pod_panel_screw.stl	upside down: top on the bed	21 g
Door stl/print/pod_door.stl	on its top edge, with a brim	1 g
Cover for screws stl/print/cover_screw.stl	on its rear end: tree supports, 14-line brim	10 g

Settings

Slice with the profiles in profiles/taz-workhorse-se: pla-0.20-standard, or pla-0.10-fine for the *_fine files. Use a brim. printdeck.yml has the settings of each part. Walls 2 mm: 4 lines of 0.5 mm. The fine panel has smaller hook gaps (0.05 mm).

pod_body_screw.3mf has support blockers on the two amp nubs: with support on, the nubs get none. The panel prints top down: its openings are on the bed. starter_plate_screw.stl (optional) has the same 4 bosses: use it to test the inserts or nuts on the cover.

Print plates

stl/plates/ has every part on one bed, in its print orientation. Each 3MF has one object for each part.

PRINTER	FILE	ORIENTATION
FDM: LulzBot TAZ Workhorse, 0.5 mm nozzle	fdm_taz.3mf	The screw parts as in stl/print/. The cover stands on its rear end: turn on tree supports and a 14-line brim.
Resin: Peopoly Phenom (first model)	resin_phenom.3mf	The resin parts with nut pockets. The body, the panel, the door and the starter plate are tipped 45 deg. The cover stands on its rear end. Lift the parts 5 mm in the slicer and add supports. stl/print/resin/ has each part alone.

Resin supports

- Body: the floor is toward the plate. Put the supports on the floor underside, off the screw holes. Sand the marks flat.
- Panel: the open side is toward the plate. The top face has no marks.
- Cover: it stands on its rear end. Put the supports on the rear edge only.
- Door: the outer face is up, tipped 45 deg. The supports go on the inner face and the ears.
- The resin check found no cups (trapped resin) in these orientations.

Resin fits

- The resin parts use glued nuts in place of inserts (page 4).
- Door gap 0.12 mm, amp slot gap 0.15 mm (a best guess).
- Print the starter plate first to test the nut pockets.
- Wash the parts in IPA and post-cure them before the fit test.

2

Inserts or nuts

Six bosses hold the threads for the screws: 4 in the floor for the cover screws, and 2 on the speaker posts for the panel screws. An FDM print (PLA) uses M3 heat-set inserts. A resin print uses M2.5 nuts with superglue.

	FDM PRINT (PLA): HEAT-SET INSERTS	RESIN PRINT: GLUED NUTS
Parts	stl/print/: pod_body_screw, pod_panel_screw, cover_screw	stl/print/resin/, or the resin plate
In the bosses	6 x M3 heat-set insert. Holes 3.6 mm, 3.9 mm deep: too big for M2.5 inserts. Bosses 7 mm.	6 x M2.5 hex nut, 5 mm across the flats, 2 mm thick. Hex pockets. Bosses 8 mm.
Screws	4 x M3 x 4 button or pan head (cover). 2 x M3 x 12 (panel).	4 x M2.5 x 4 button or pan head (cover). 2 x M2.5 x 12 (panel).
Tools	Soldering iron (an insert tip helps), screwdriver	Superglue, screwdriver

Heat-set inserts (FDM)

1. Put an insert on the hole of a boss.
2. Heat it with the soldering iron, and press it in straight until it is flush with the top of the boss.
3. Do the 4 floor bosses and the 2 speaker post bosses.
4. Let the plastic cool before you put in a screw.

Glued nuts (resin)

1. Put a nut flat into the hex pocket at the top of a boss, and push it to the bottom.
2. Put a drop of superglue on the edge of the nut. Keep the glue off the thread.
3. Do the 4 floor bosses and the 2 speaker post bosses.
4. Let the glue set before you put in a screw.

Floor nuts: the nut is at the bottom of its pocket. The M2.5 x 4 cover screw goes fully through it (screw tip at Y 29.2, nut top at Y 29.0). Speaker post nuts: the panel column is over the nut and holds it in its pocket.

3

Electronics and cables

Each cable below shows its two ends. Some have one plug at one end and many at the other: match each end before you plug it in.

Amp cable

`1 plug → 5 sockets`

Comes with the NULLLAB NS4168 amp: one 5-pin PH2.0 plug at the amp, five single female Dupont sockets at the other end. The wires are bonded flat. Keep the cable on edge: do not twist it.



BiBoard jumpers

`3 single jumpers` `female → male`

The female end goes onto the Pi pin. The male end goes into the BiBoard socket, which is female.



Mic jumpers

`6 single jumpers` `female → female`

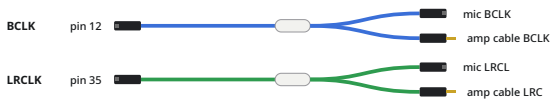
One per mic pin. BCLK and LRCL end in the free female ends of the splices; cut them to length when you make the splices.



Splices

`1 → 2`

Each clock signal feeds two boards. Make a Y from three jumper ends: solder, then cover with white heat shrink.



Speaker lead

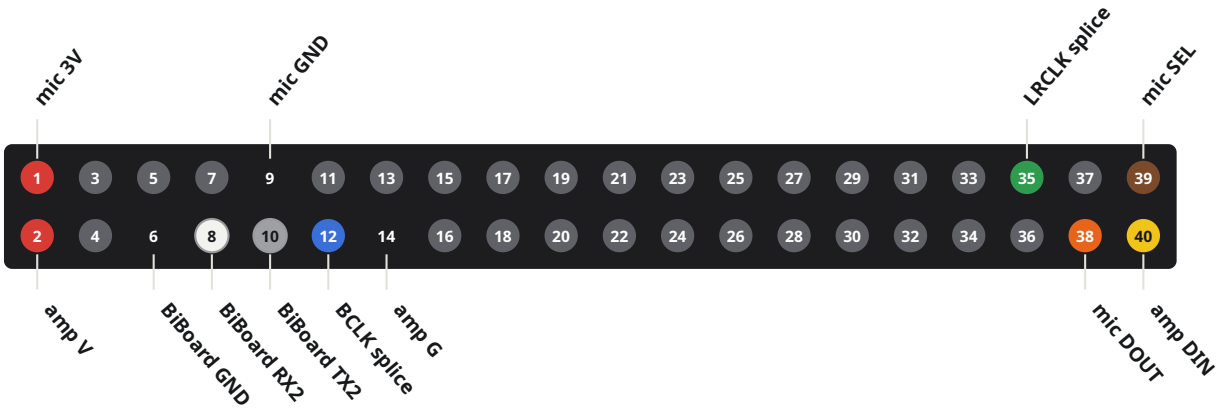
The 2-wire lead of the speaker ends in a 2-pin PH2.0 plug: it goes into the rear end of the amp.

4

Pin maps and hardware

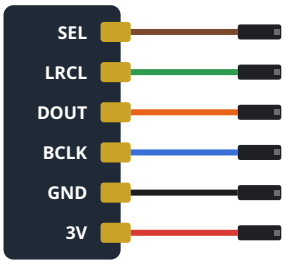
Every pin in use, in the color of its wire.

Pi Zero 2 W, 40-pin header



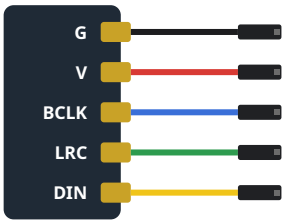
Pin 1 is at the SD card end (rear). Odd pins: inner row (top). Even pins: board edge (bottom).

Mic (as mounted)



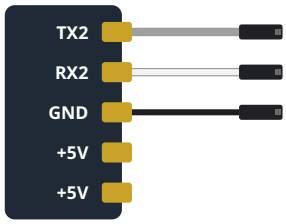
SEL at the top, 3V at the bottom

Amp I2S plug



G at the top (the amp stands upright)

BiBoard socket



male ends in; +5V stays empty

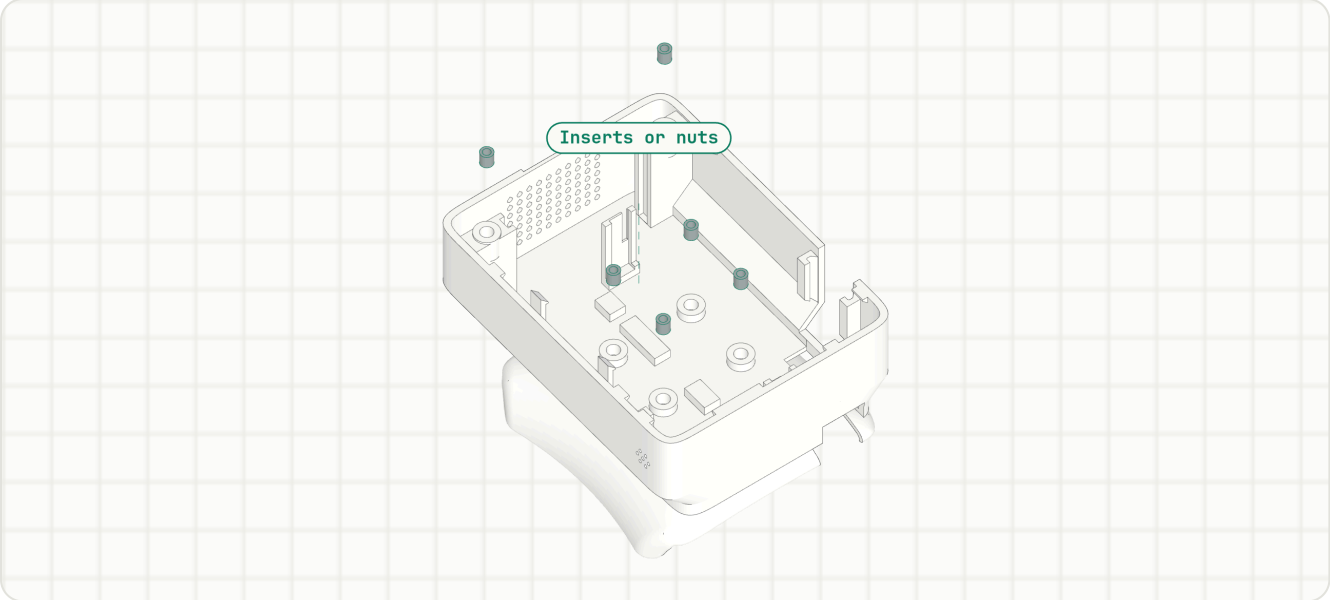
TX and RX cross: BiBoard TX2 goes to Pi RXD (pin 10), BiBoard RX2 to Pi TXD (pin 8). SEL to GND puts the mic on the left channel. The PiSugar S powers the Pi through its contact pins.

Hardware

ITEM	COUNT	USE
Female-to-male Dupont jumper, 10 cm	3 + 2	BiBoard serial link (3) and the amp ends of the splices (2)
Female-female Dupont jumper, 10 cm	4 + 2	mic (4) and the Pi ends of the splices (2)
Female-female Dupont jumper, 15 cm	2	mic GND and 3V
Heat shrink, white	2	splice sleeves
Glue (a dab)		amp
Screws, and inserts or nuts	6 + 6	M3 (FDM) or M2.5 (resin): page 4

5

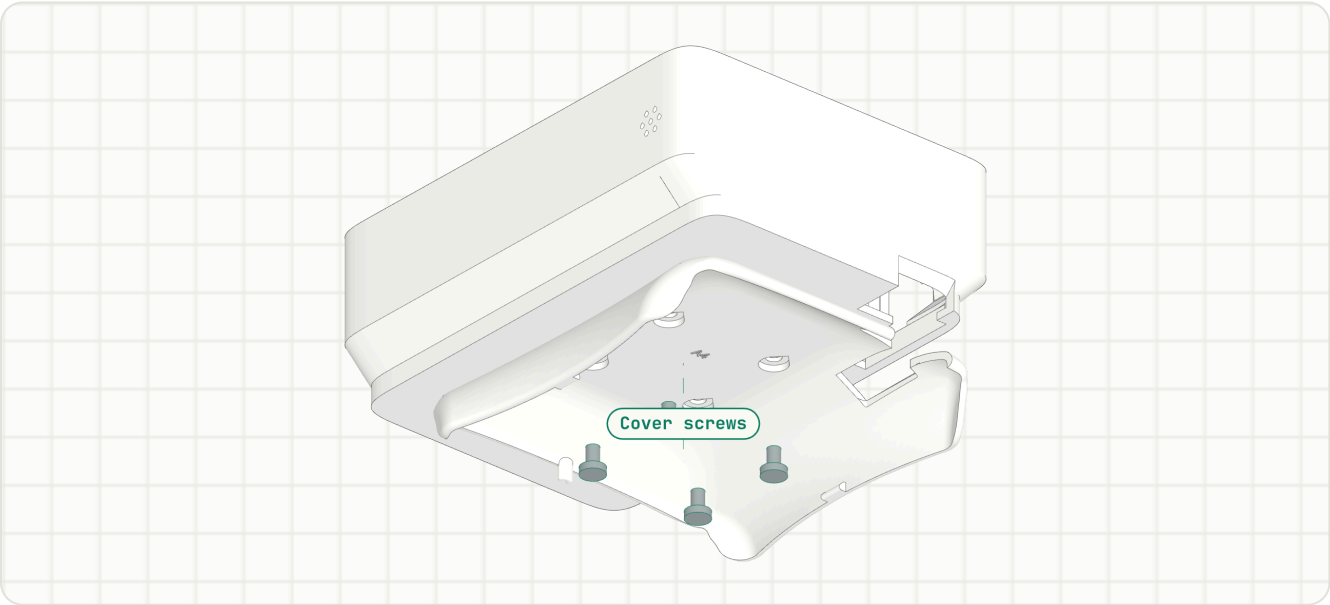
Assembly



1 Inserts or nuts

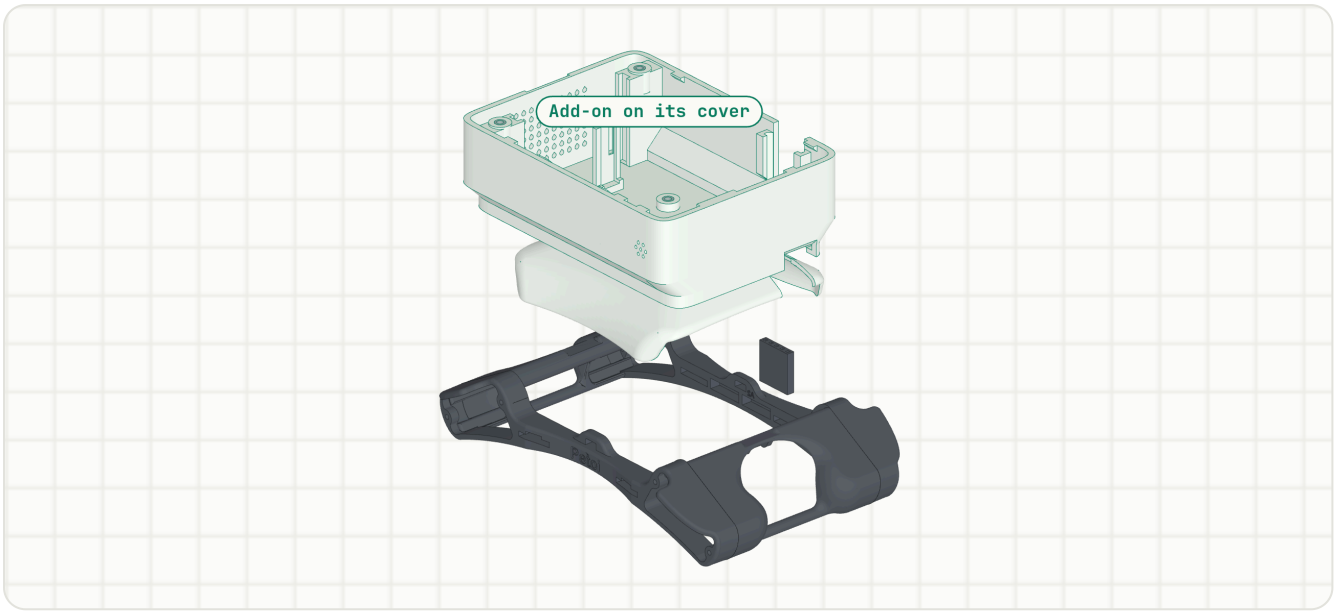
1. Put an insert or a nut into each of the 6 bosses: 4 in the floor, 2 on the speaker posts.
2. FDM (PLA): press each M3 heat-set insert in flush with a soldering iron.
3. Resin: put an M2.5 nut into each hex pocket, with a drop of superglue on its edge.

> Page 4 has the hardware, the tools and the steps for each way.



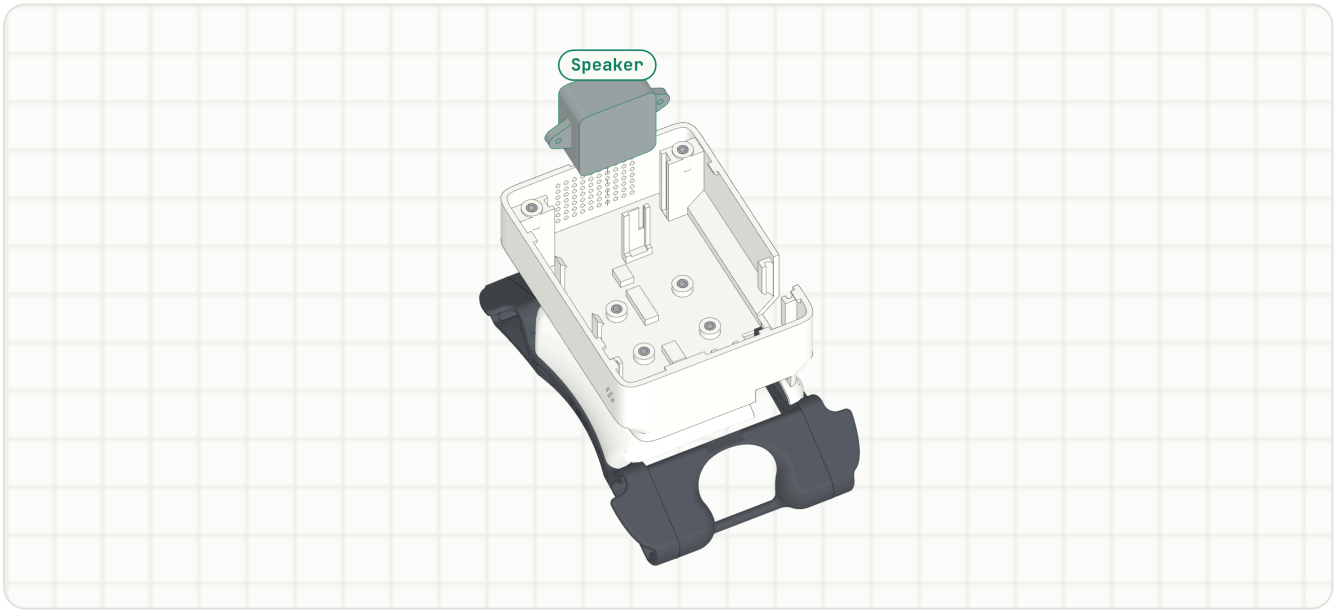
2 Screw the body to the cover

1. Put the empty body on the cover for screws, the grille to the rear.
2. From inside the cover, put a screw up through each of the 4 rings into the insert or the nut: M3 x 4 (FDM) or M2.5 x 4 (resin).
3. Tighten the 4 screws. The floor is on the rings.



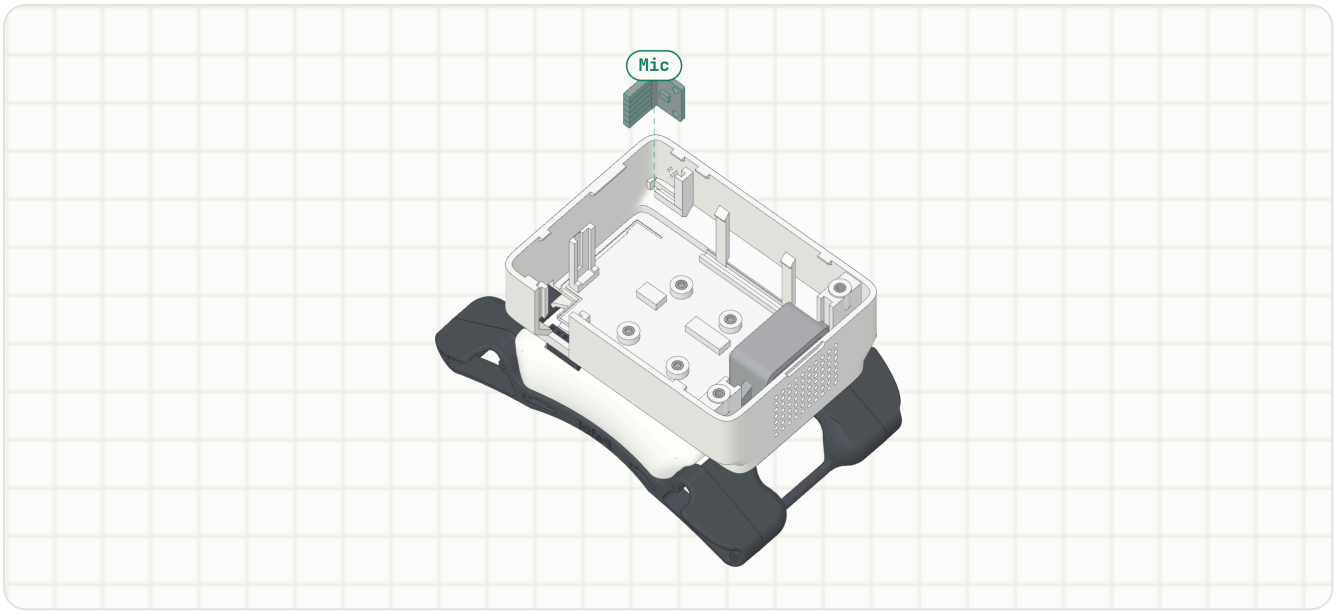
3 Clip onto the robot

1. Remove the stock back cover of the Bittle X.
2. Clip the cover and the body onto the robot as one unit, in place of the stock cover.
3. The BiBoard header comes up through the opening in the cover and the notch at the front of the left well.



4 Speaker

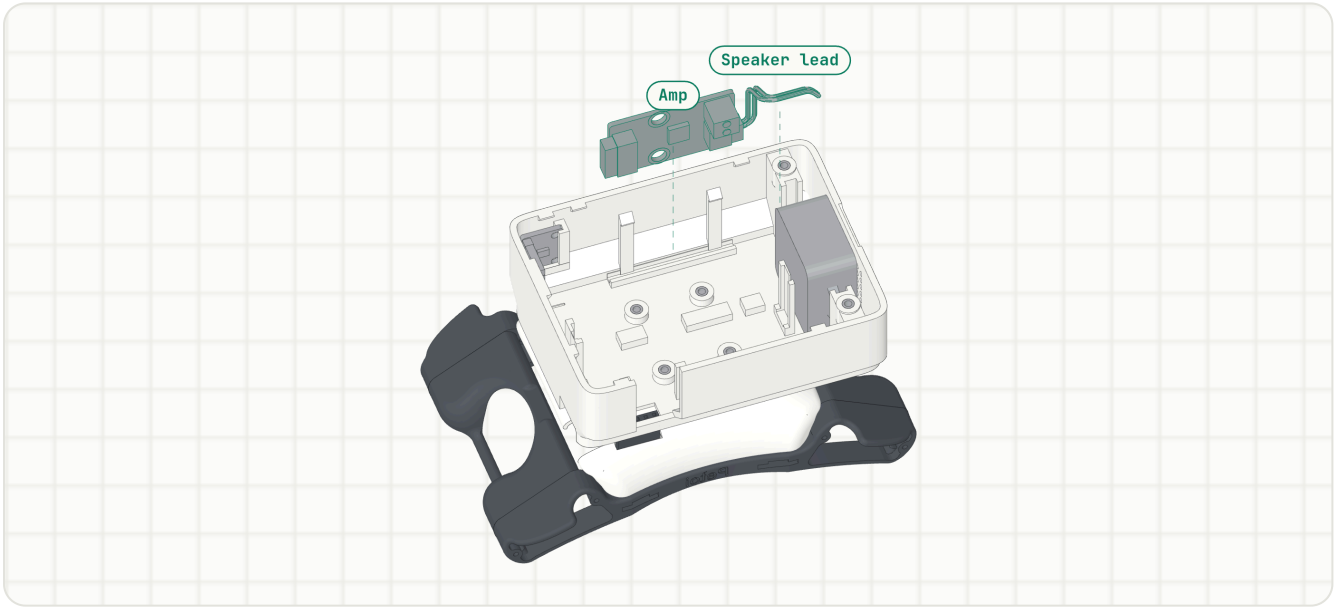
1. Push the two wings of the speaker down into the side grooves at the rear.
2. The speaker body goes to the rear, toward the grille.



5 Mic

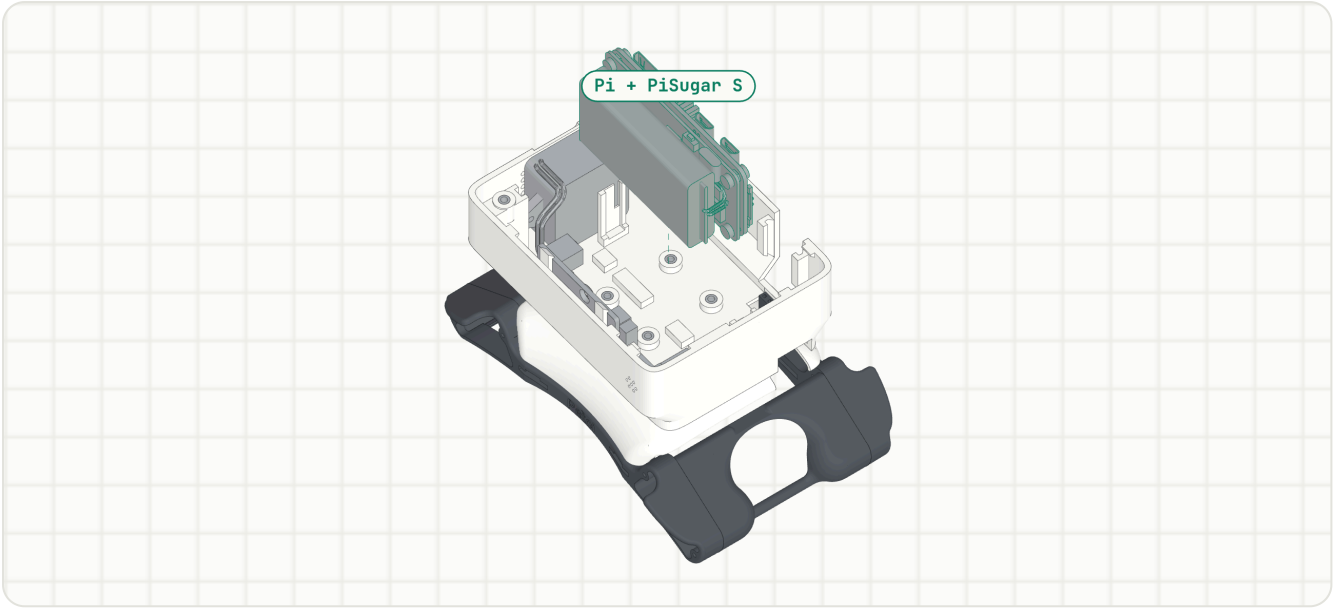
1. Plug the six mic jumpers onto the mic header. Use 15 cm jumpers on GND and 3V.
2. Drop the mic into the slots in the post and the ledge on the right wall, near the front.
3. The port faces the wall. The header is at the front edge, the plugs point into the body.

> The board sits in both slots. The post, the ledge and a stop hold it.



6 Amp and speaker lead

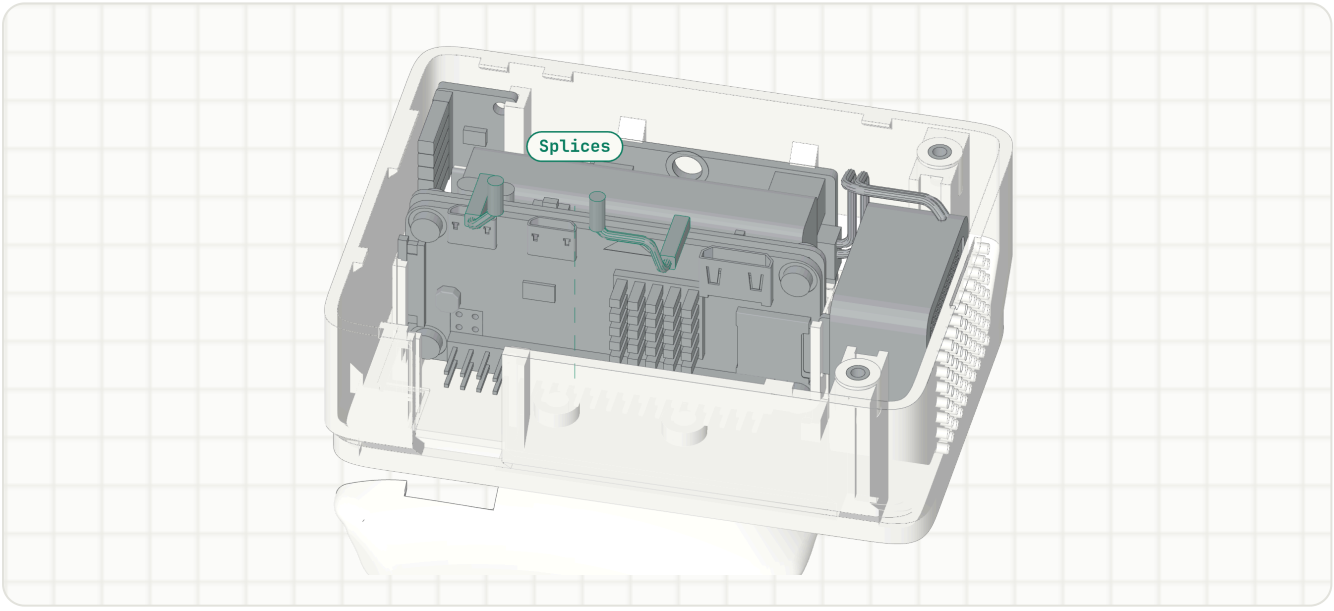
1. Plug the speaker lead into the 2-pin connector at the rear end of the amp.
2. Plug the flat amp cable into the 5-pin connector at the front end. G is at the top.
3. Put the bottom edge of the amp into the slot along the right wall, parts to the inside.
4. Tilt the top edge under the two nubs. Set it with a dab of glue.
5. Two forks under the panel hold the top edge later.
6. Let the amp cable hang out of the top. It goes in after the BiBoard jumpers.



7 Pi + PiSugar S stack

1. Set the PiSugar AUTO switch first: it is under the cell.
2. Push the stack down into the grooves of the two end posts until it stops.
3. The ports are up, the 40-pin header down and toward the left wall, the SD card at the rear.

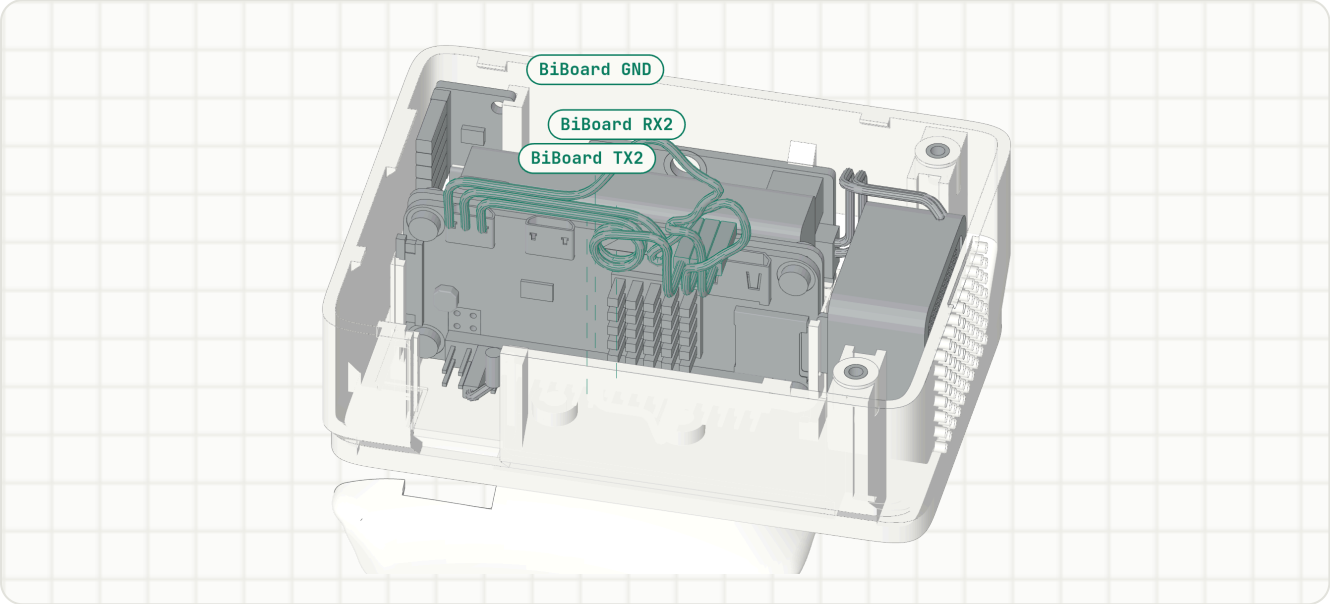
> If the stack is not fully down, the panel does not slide on later.



8 Splices (BCLK and LRCLK)

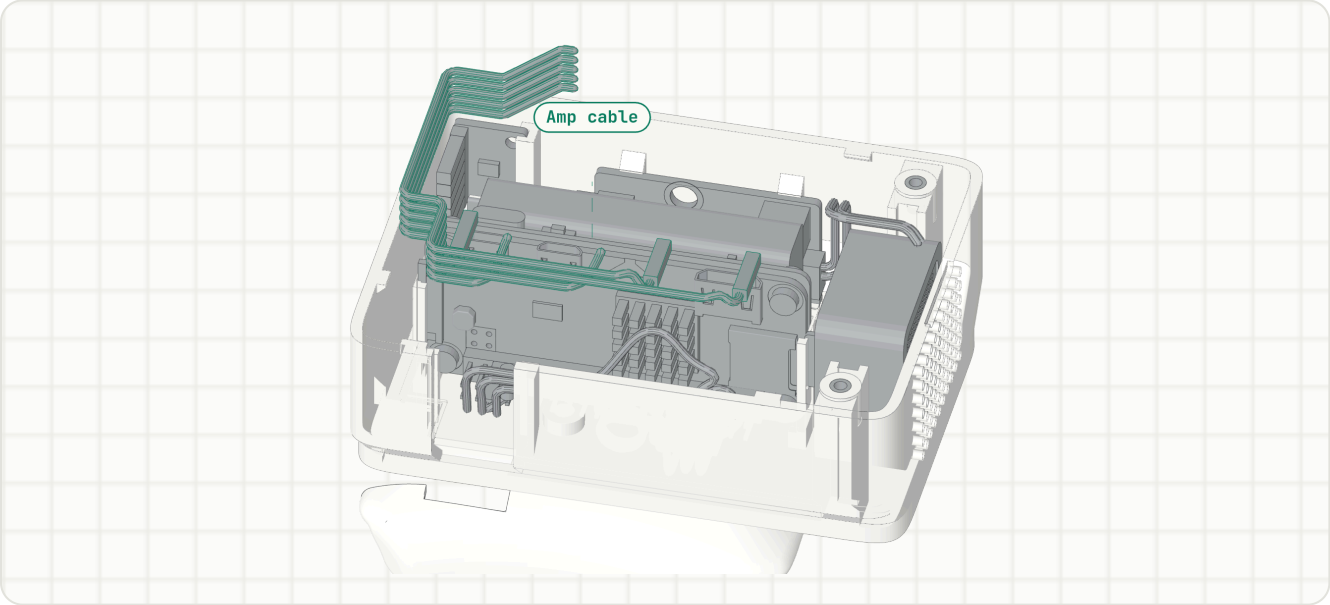
1. Plug the Pi end of each splice: BCLK on pin 12, LRCLK on pin 35.
2. The LRCLK sleeve stands beside the BiBoard plugs, the BCLK sleeve at the middle of the left well.

> Each splice is a Y: one Pi end, one mic end, one amp end. See the cable cards.



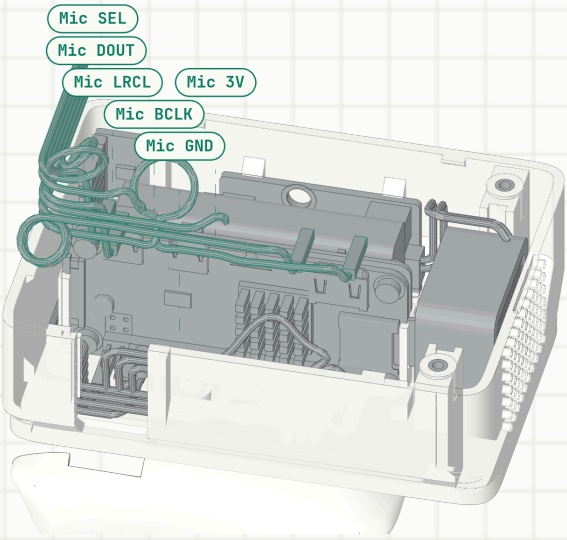
9 BiBoard jumpers: Pi ends

1. Push the female end of each BiBoard jumper onto the Pi: TX2 (gray) on pin 10, RX2 (white) on pin 8, GND (black) on pin 6.
2. Lay the excess of each one in a loop by the left wall.
3. Leave the male ends free. They go onto the BiBoard last.



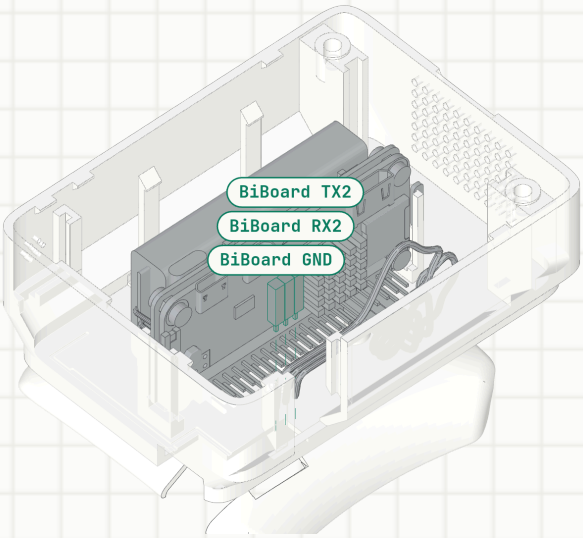
10 Amp cable

1. Lay the flat amp cable on edge, as it leaves the plug, G on top. Do not twist it.
2. It steps down, runs forward to the front wall and across under the mic plugs. Past the front post it turns rearward, runs across behind the front-left notch, and turns rearward along the left well.
3. Peel off each wire at its own height: DIN onto pin 40, LRC and BCLK into the free sockets of their splices, G onto pin 14, V onto pin 2.



11 Mic jumpers

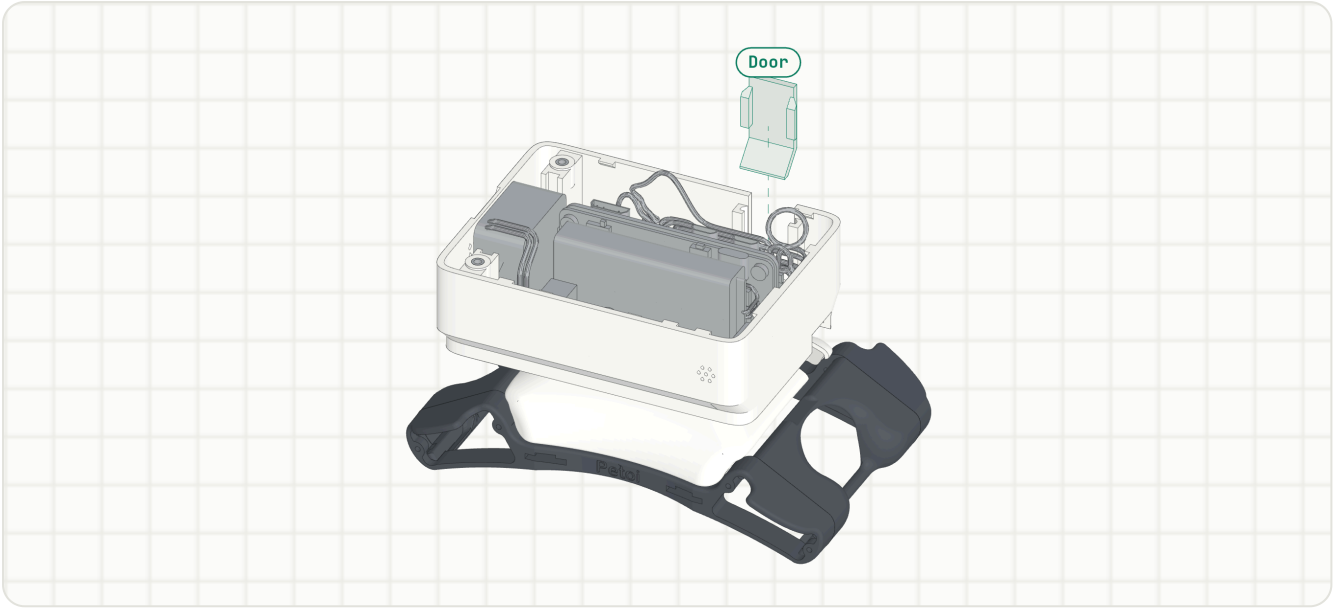
1. Run the mic jumpers straight across behind the front wall, then rearward along the left well.
2. Plug them from the lowest up: 3V on pin 1, GND on pin 9, BCLK and LRCL into their splices, DOUT on pin 38, SEL on pin 39.



12 BiBoard ends

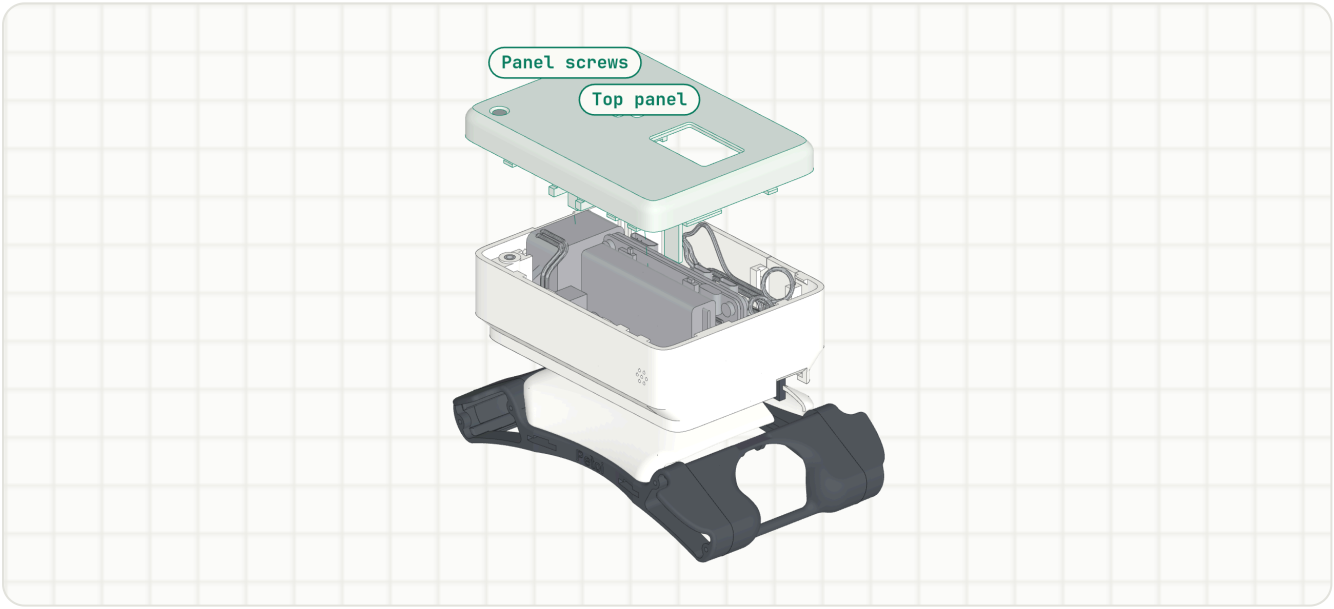
1. Push the male ends of the three BiBoard jumpers into the BiBoard socket: TX2, RX2, GND.
2. Reach the plugs from the side, through the door opening in the left wall.
3. The two +5V sockets stay empty: the PiSugar powers the Pi.

> The other wires are not shown here, for a clear view into the front-left corner.



13 Door

1. Slide the door down into the two grooves of the opening in the left wall.
2. It stands on the wall under it. The panel holds it down.



14 Panel

1. Put the panel down 5 mm behind its place, and slide it forward.
2. Put the two panel screws down through the panel into the speaker posts: M3 x 12 (FDM) or M2.5 x 12 (resin).
3. The PiSugar power switch is in the HDMI opening, beside the plug. Move it with a pen tip or a toothpick.
4. Press the PiSugar button through the large opening with a fingertip or a pen tip.

> To open: take out the two screws and slide the panel rearward.

6

Wire table

Every wire, from end to end. Lengths are between the housings, as modeled; a jumper longer than the route lays its excess in the loop shown.

FROM	TO	GROUP	LENGTH	EXCESS
BiBoard TX2	pin 10	uart	100 mm	2-turn loop
BiBoard RX2	pin 8	uart	98 mm	arch
BiBoard GND	pin 6	uart	98 mm	arch
amp G	pin 14	amp	116 mm	
amp V	pin 2	amp	130 mm	
amp BCLK	BCLK splice	amp	109 mm	
amp LRC	LRCLK splice	amp	95 mm	
amp DIN	pin 40	amp	81 mm	
BCLK splice	pin 12	splice	16 mm	
LRCLK splice	pin 35	splice	7 mm	
mic 3V	pin 1	mic	148 mm	1-turn loop on a 5 mm stem
mic GND	pin 9	mic	149 mm	2-turn loop
mic BCLK	BCLK splice	mic	76 mm	cut to length
mic DOUT	pin 38	mic	97 mm	1-turn loop on a 5 mm stem
mic LRCL	LRCLK splice	mic	62 mm	cut to length
mic SEL	pin 39	mic	99 mm	1-turn loop
amp speaker +	speaker +	speaker	43 mm	
amp speaker -	speaker -	speaker	44 mm	

7

Checks, opening and removal

What the model checks

- No printed part and no board touch, except where a fit is meant to touch.
- Every part goes in from the top, in the order of this guide.
- No two wires pass through each other. No designed bend is tighter than 2 mm.
- The head turns from -85 to 85 deg without contact.
- The cover and the body go straight down onto the robot together. Each screw is clear of the boards.
- Mass 114 g; center of mass 8.8 mm from the robot center.

Open the panel

Take out the two panel screws, slide the panel 5 mm rearward and lift it.

Remove the add-on

Open the panel. Lift the door out of the left wall. Pull the three BiBoard plugs off from the side. Unclip the cover from the robot: the body comes off with it. To take the body off the cover, take out the 4 screws from inside the cover.

Change the SD card

Open the panel and lift the speaker out of its grooves; the card is at the rear end of the Pi.

Files

stl/print: the parts to print (stl/print/resin: the resin parts). stl/plates: the print plates. cad/: STEP files. README.md: the full reference. Designs CC BY 4.0, code MIT; see LICENSE.md and ATTRIBUTION.md.